FPGA

Stimulus & Sampling Pulse Timing Generation

U5 — LFE5U-12F-6BG256C

Pulse Amplifier

U2 — PI5HDX12311

Signal Splitter

Couples incident stimulus into mesh, reflections into sampler

Sampling Bridges

BAT15-04W

Clip Lines

Place clip line close to diodes

Low-Speed Amplifiers

U13A — OPA4354

U13B

U13C

U13D

L/R Connector Mux

U10 — TMUX1574

Termination Mux & Crossbar

Selects termination, and flips mesh send/return

U4 — TMUX5446

U12 — TMUXHS4446

Sub-Mesh Output Mux

U8 — PI3WVR41310

TODO: ESD diodes here?

U9 — PI3WVR41310

J1 — Conn_01x30_MountingPin

J2 — Conn_01x30_MountingPin

Microcontroller

U15 — STM32U575RGTx

Y1 — Crystal_GND24_Small

Debug USB

J5 — USB_C_Receptacle_USB2.0_16P

U16 — USBLC6-4SC6

Debug connector

J3 — Wuerth WR-WST Redfit IDC

PiHAT+ Connector

HAT+ ID EEPROM

U11 — M24C32-FMN

J4 — Raspberry_Pi_4

Power Supplies

5V input, outputs: 3V3D (digital), 3V3A (analog, low noise), 1V8 (digital), 1V2 (digital)

U7 — TMT7003C

U8 — TLV74333P08VR

U5 — TLV74312P08VR

U17 — TPS512000RC

let's sense some pillows~! ♡

✦ センサー回路図 ✦

yanartas-pillowbox-sensors

~ the pillowbox sensor board ~

sheet ♡ 1 / 1

KiCad 10 · paper A1 · drawn with love ♪